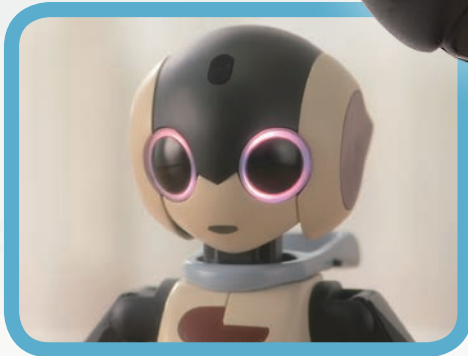
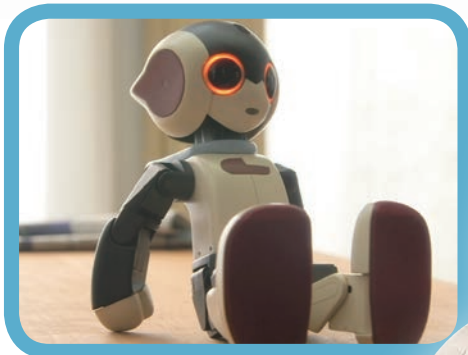


Build your own **Robi**

Pack **06**



Stages 19-22:
Continue building
Robi's left arm





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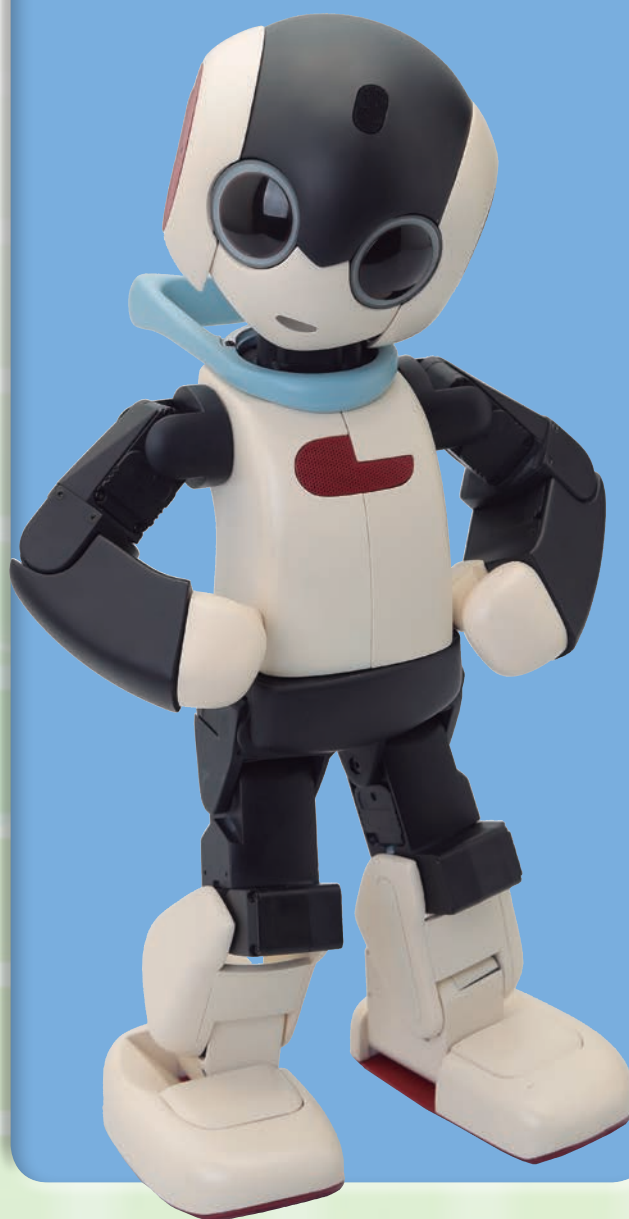
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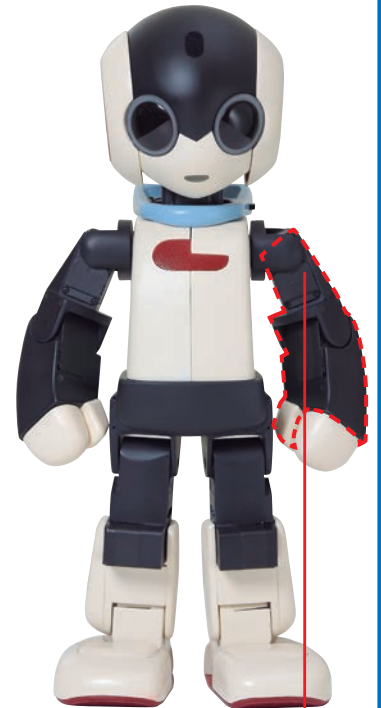
STAGE 19: SET THE ID NUMBER OF ROBI'S LEFT UPPER ARM SERVO

In this stage, you will set the ID number of the servo that will act as Robi's shoulder joint, moving his left upper arm up and down, before fitting the part onto the left forearm frame, ready for use.

The work in this assembly will be similar to that of Stage 16, when you wired and set the ID number of the servo that will move Robi's left forearm. Even so, make sure you follow the assembly steps closely, as it can be difficult and, for the electronic parts such as the servo cables, even

damaging, to re-open the assembly should you need to go back and perform the steps again.

Similarly, when setting the servo's ID, be absolutely sure you choose the correct number, as Robi may not work properly if the part is set to the wrong number when operated.



PARTS TO BE ASSEMBLED

1



YOUR PARTS

1 Servo motor
(for left upper arm)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Servo cable and left arm assembly (Stage 18)



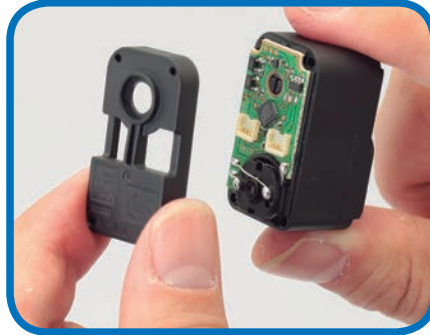
Neck stand (Stage 8)

CONNECTING THE SERVO CABLE

BE CAREFUL!



Try to avoid touching the circuit board or electronic components when you remove the cover from the servo.

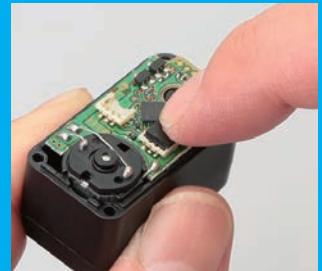
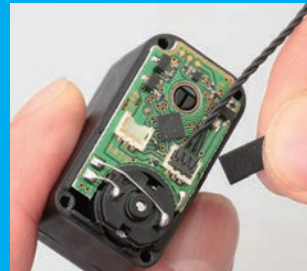


1 Use a screwdriver to undo the four screws securing the cover of the servo, and take each one out.

2 Carefully lift the cover away from the other side of the servo.



TIP!



3 Take either end of the connector cable saved from the previous stage and carefully fit it in one of the two sockets circled in red. It does not matter which one. Gently push it until it locks in place.

If you chose not to fit the protective pad previously, you can fit it now, which will help to secure the connector in place. Do this very carefully, avoiding any direct contact between your fingertip and the exposed electronic components on the servo's open board.

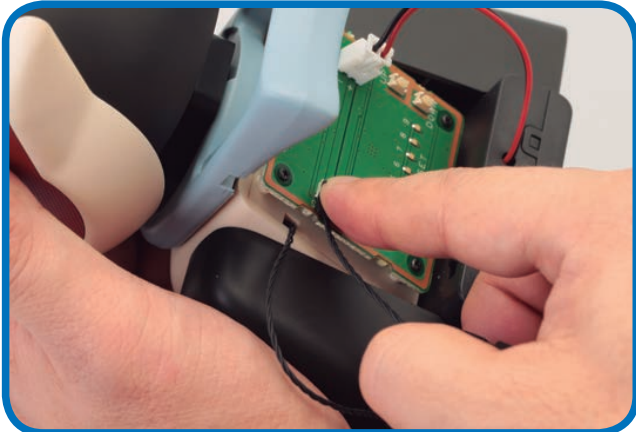


4 Carefully feed the free end of the cable up through the circular hole in the servo cover. Make sure that the cover is the right way up.

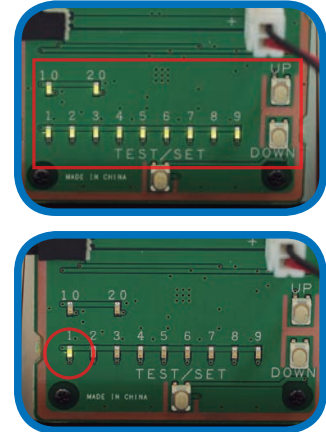


5 Fit the cover onto the servo casing, then turn the servo over and insert two of the screws you removed in Step 1. Tighten them lightly, as you will be removing them again in this assembly.

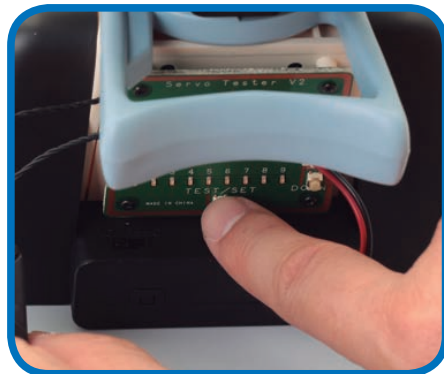
TESTING THE SERVO



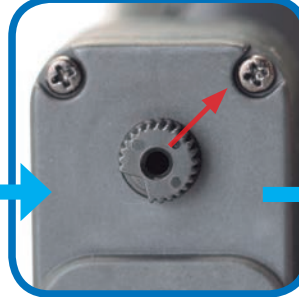
- 6** Take the neck stand and unplug the black servo cable from the socket on the left of the test board. Plug the cable leading to this stage's servo in its place.



- 7** Turn the test board to ON. All the LEDs should blink twice, then turn off to leave only number 1 lit.



- 8** Now press the TEST/SET switch briefly to run the servo test.

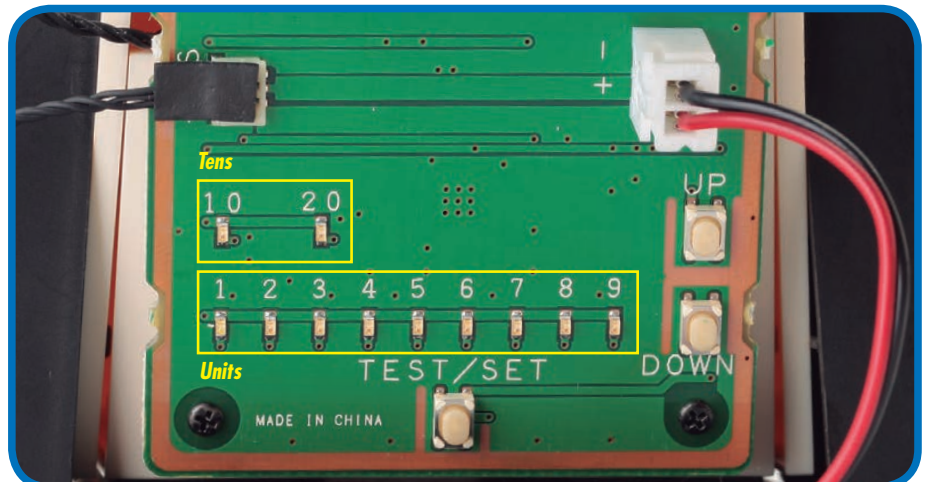


- 9** The shaft at the top of the servo casing should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. After you have run the test, turn off the power switch on the battery pack.

HELP!

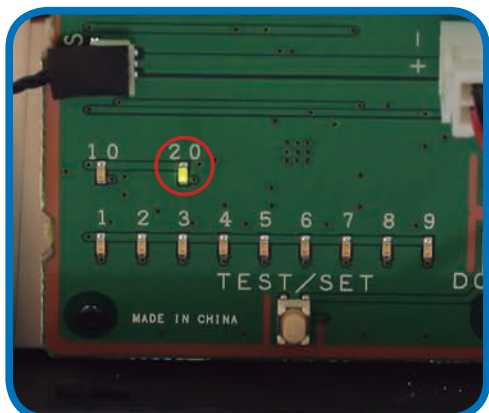


If the LEDs flash rapidly when you run the test at Step 14, you have a faulty connection. Double-check the servo cable's connection to the socket on the test board, **TURNING THE POWER PACK OFF BEFORE YOU DO**. If the light still flashes, carefully unscrew the servo housing and check that the internal connector plug is in place. If you pressed the wrong button, reset the board by holding down the TEST/SET switch for several seconds.



- 10** Re-familiarise yourself with the test board. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 21, as you did for the left forearm servo, you'd press UP 20 times. This will light up the 20 LED and the 1 LED. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.

SETTING THE ID NUMBER

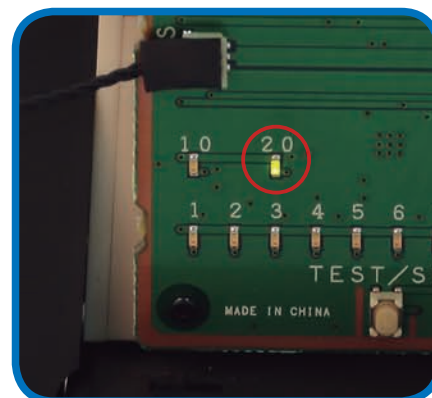
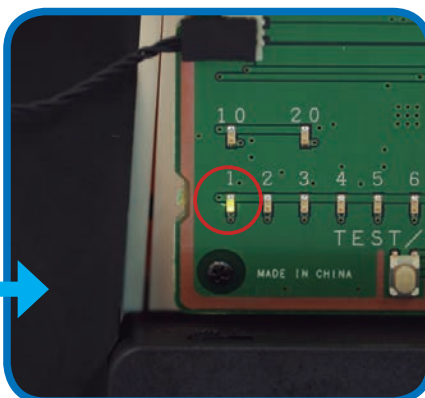


ID=20

11 The servo that will move Robi's upper arm is ID=20, so to set the ID number to 20, press the UP button 19 times. The 20 LED alone should remain lit once you have done this.

12 Complete the setting by holding the TEST/SET button down for a few seconds. The 20 light should flash, confirming the ID number.

CHECKING THE ID NUMBER



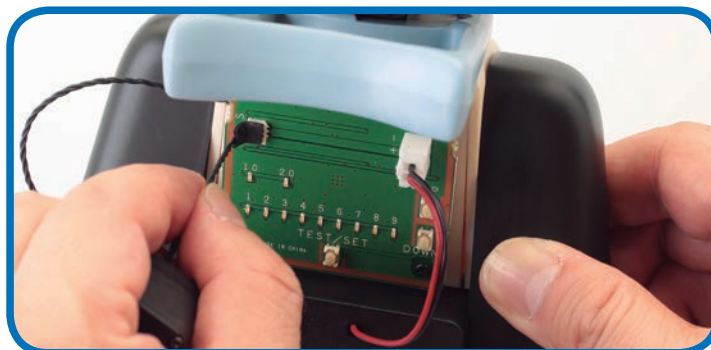
13 To begin checking the ID number, leave the servo connected, switch the power OFF and then back ON. Only the 1 LED should light up.

14 To check the ID number, press TEST/SET. Now the 20 LED should light up.

TIP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake and set the wrong ID number, you can reset it by going through the setting procedure again.



15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to be fitted to Robi's arm.

FITTING THE SERVO



- 16** Remove the two screws fitted temporarily in Step 5 and discard the cover. Now line up the open servo with the left upper arm frame. The servo cable will fit through the circular hole at the top of the frame, as shown.



- 17** Holding the servo in position with its cable sitting in the upper arm frame's hole, carefully attach the free cable (leading from the forearm servo) to the unused connector socket on it. This can be a bit fiddly, so take care and be gentle as you do this.

HANDLE WITH CARE!



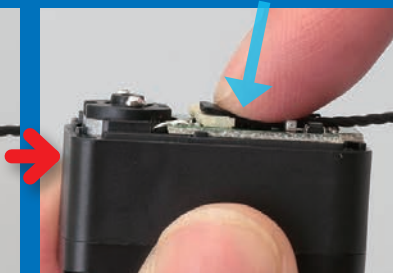
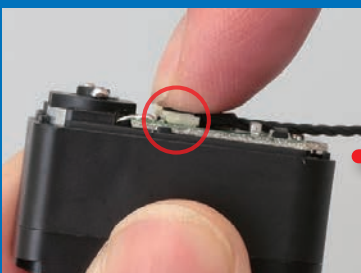
The servo cables carry the electrical signals that operate Robi's motors. It is essential that each one is connected properly, and not placed under strain. If a cable is damaged, you may find that one of Robi's motors won't work at all, or stops working after a time.

Check the contacts

Make sure that each connector on the end of a cable is pushed all the way home into its socket. Otherwise, you may find that it works loose in use, due to the movement of the servo.



Even if the connector appears to have made contact, you may find that the base is not properly engaged with the socket, and can work loose.



Ensure that the connector is pushed all the way forwards and all the way down.



- 18** Making absolutely sure that neither of the servo cables is caught between the servo's body and the upper arm frame, and that each cable's connector is securely and safely attached to its connector (see box on previous page), join the servo to the arm frame, as shown. It should sit flush to the frame.



- 19** Using the four long screws removed previously, fasten the servo to the upper arm frame, again making absolutely sure neither of the cables is caught in the gap between the parts.

There are two small nibs moulded into the inside of the upper arm frame (circled). These should be avoided when laying the servo cables inside the subassembly, to prevent the risk of their getting pinched by them when you close the parts. Laying the cable centrally within the space should prevent this happening.



TIP!



By now, you will have accumulated a number of components, spare screws and subassemblies in progressive stages of completion. It will be useful to store these in an orderly manner, and for this a compartmentalised tool or hobby box will be ideal, like the one shown above.

Note: Do not connect the servos linked in this stage to the test board in any way from now on, as this may affect their individually programmed ID numbers.

Assembled left arm

Robi's left arm is nearly complete, ready to be joined to his body.



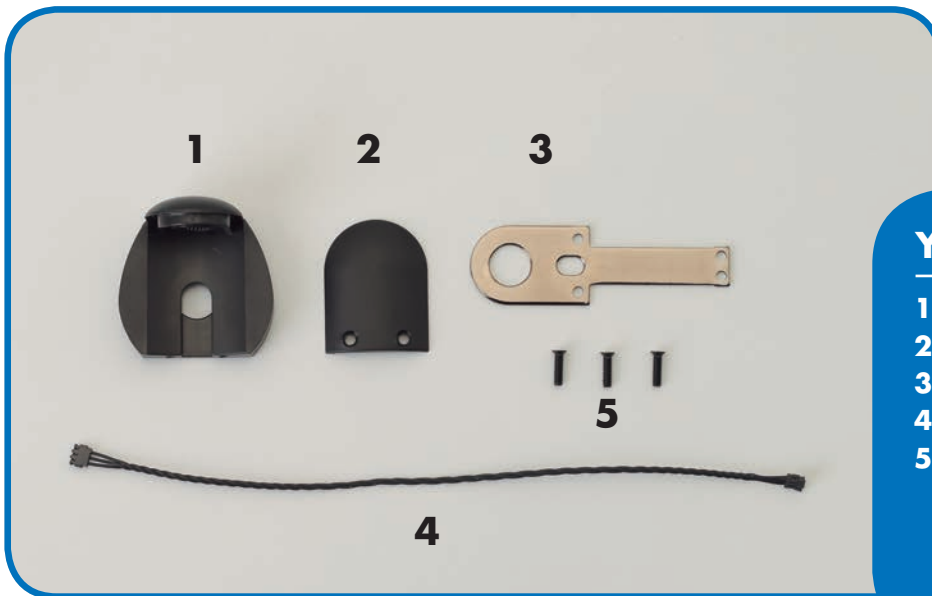
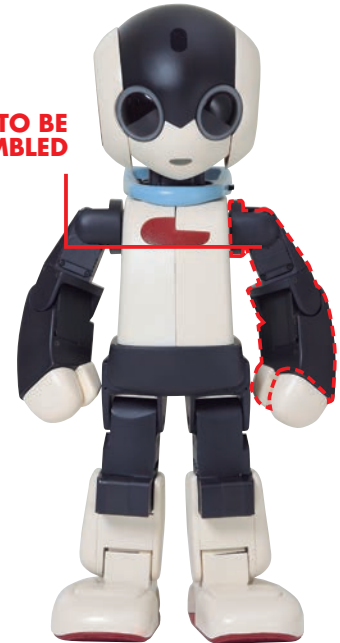
STAGE 20: ADDING ROBI'S LEFT SHOULDER JOINT

In this assembly you will fit the left arm built over the previous stages to its shoulder frame, mirroring the work you did for the right shoulder in Stage 12.

By the end of this stage, Robi's left arm will be complete, and ready to be attached to his body at the shoulder over future stages. Once Robi's arm is prepared, you will add protective pads to the connectors of another of his servo cables, so as ever care must be

taken when handling these electronic parts. Though this work in this stage is very similar to that of Stage 12, still be absolutely sure to follow the instructions closely, as it will be difficult to go back and change the completed assembly later.

PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Shoulder frame
- 2 Shoulder panel
- 3 Shoulder joint
- 4 Servo cable (135mm long)
- 5 M2 x 7mm countersunk screws (x 3)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

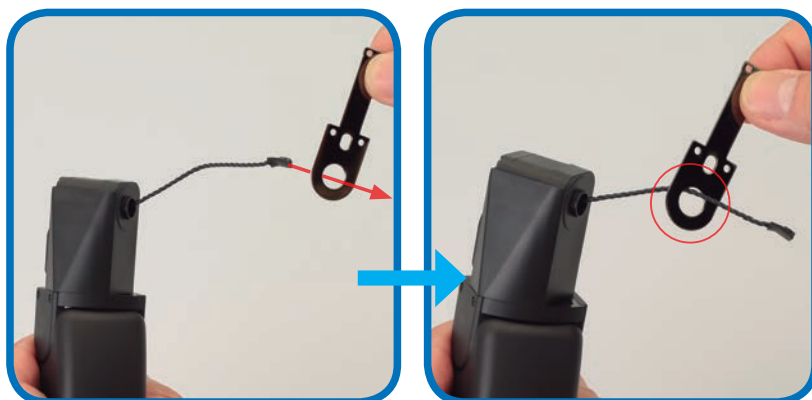


Left arm assembly (Stage 19)



Protective pad sheet (Stage 3)

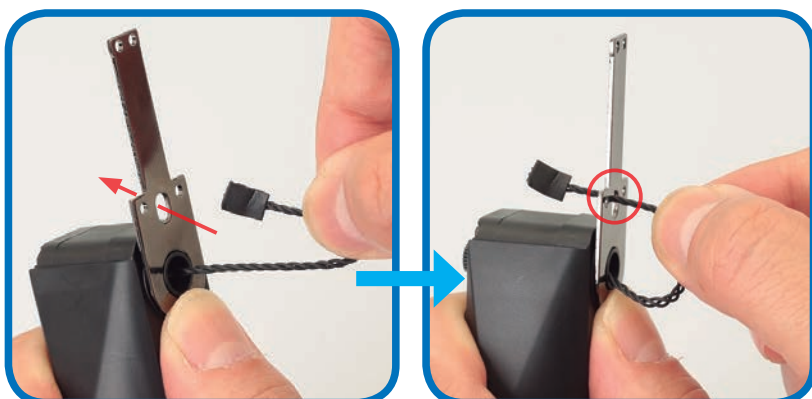
WIRING THE SHOULDER JOINT



- 1 Feed the servo cable coming out of the upper arm frame through the hole in the shoulder joint. It does not matter which way round the joint is positioned.



- 2 Run the shoulder joint along the cable and fit its circular hole around the raised opening of the upper arm frame.



- 3 Holding the parts in place as shown, feed the tip of the servo cable over and through the indicated hole in the shoulder joint.



- 4 Keep holding the parts together with your finger and thumb, then pull the cable gently so that it is taut.

ADDING THE SHOULDER FRAME



- 5 Feed the servo cable through the indicated hole in the shoulder frame, orientated as shown above, and bring the parts together.



- 6 Join the parts by placing the 'D-cut' hole in the frame over the corresponding servo shaft (red arrow). See the Close Up section on the next page for more information on this.

THE 'D-CUT'

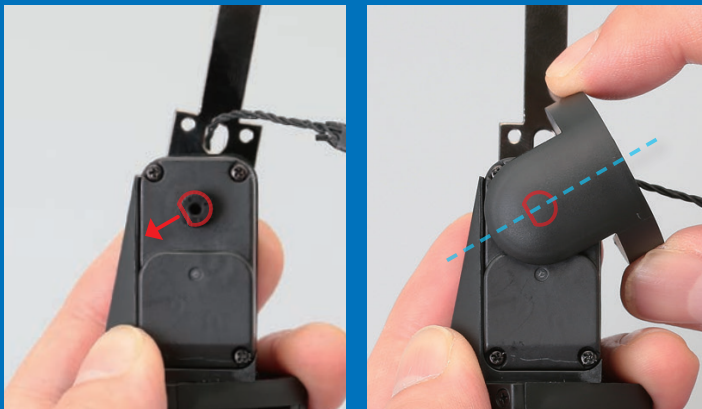
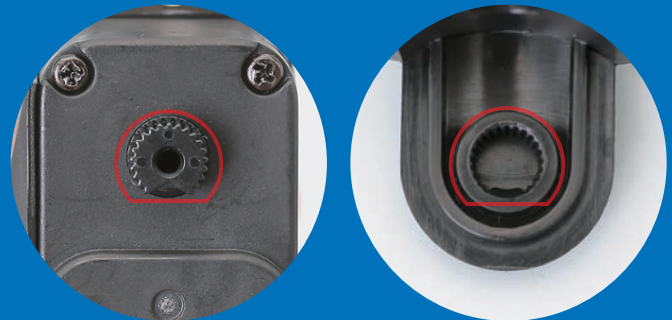


The round shafts of Robi's servos are moulded with a small notch in one side, creating a flat edge (resembling the letter 'D') that fits into a correspondingly shaped recess on the part to be joined. This ensures that the parts can be set correctly when in the neutral position.

The 'D-cut'

Left photo: the servo's shaft is not completely circular, but has a notch in its ribbed edge, creating a straight section of outline.

Right photo: the inside of the corresponding joint mirrors these ribs, as well as the straight edge created by the shaft's notch.



Restoring to the neutral position

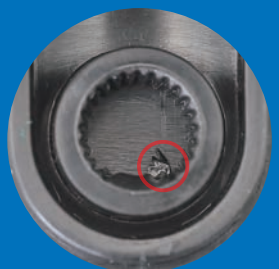
If something goes wrong when performing a test of one of Robi's servos, for example if the power is disconnected, and the servo's shaft stops turning and remains at an angle, you will need to reset the servo to its neutral position. To do this, identify the notch in the servo shaft (left photo) and place the part it will operate over it at the same angle (right photo). Then, very gently rotate the part back to the neutral position, with the servo shaft's notch facing downwards.

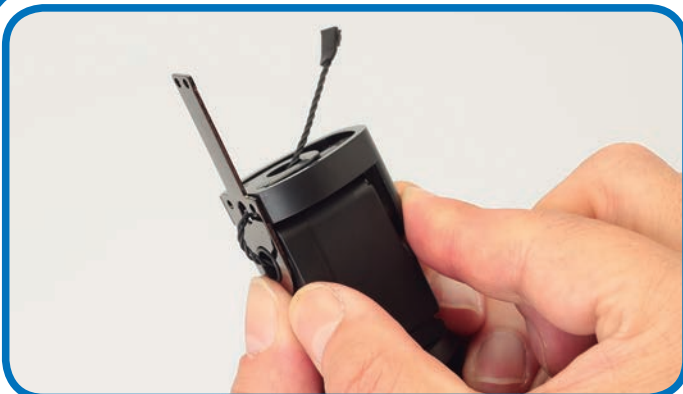


Neutral position

The advantage of using a D-cut shaft and joint set-up is that it makes it very easy to assemble Robi's servos and accompanying parts correctly. When the parts are aligned correctly, as in the far left photo, they will fit together easily, while if they are not orientated as they should be, they will struggle to fit together and result in an off-centred neutral position (near left). If this happens, do not force the part, and realign it with the servo shaft until it fits easily.

Warning: applying too much force when pressing a part over the servo's shaft can result in damaging both parts.



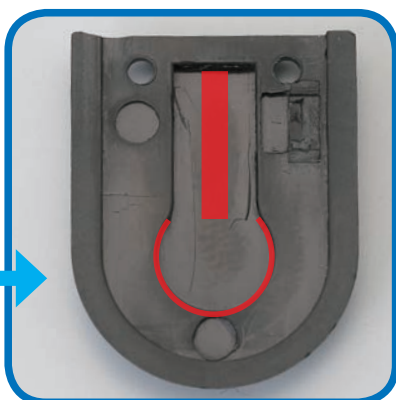


- 7** Once you have the parts aligned as they should be, press them together tightly. Be absolutely sure the parts are sitting correctly, as joining them when off-centre will damage them.



- 8** To prevent the servo cable slipping out of position, pull it again gently so that it is taut.

SEALING THE SHOULDER

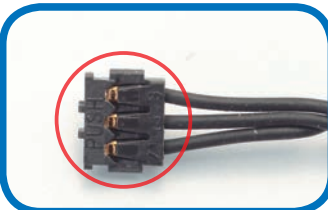


- 9** Place the shoulder panel over the shoulder joint, keeping the circled screw holes aligned. The red line in the right photo shows the groove in the shoulder panel that will take in the servo cable, while the curved line indicates the area that will accommodate the servo's raised cable hole.



- 10** Once in place, secure the parts using two M2 x 7mm countersunk screws.

SEALING THE SERVO CABLE



- 11** As you have done previously, apply a protective pad to the side of this stage's servo cable with PUSH marked on it (circled). Repeat for both ends of the cable.

Assembled left arm and servo cable

This stage is complete. Robi's left arm is assembled and wired, ready to be joined to his body in future stages.



STAGE 21: PREPARE ROBI'S LEFT SHOULDER SERVO

In this assembly you will test and set the servo that will operate Robi's forward and back arm movement, ready to be fitted into his left-side body frame in coming stages.

This stage sees you working with servo motors again. The servo supplied with this stage is ID=19, as listed on the diagram in Stage 8, and will generate the forward and

backward motion for Robi's left arm, replicating the work of a human's shoulder. Once it is complete, the servo itself will be set into his body frame, out of sight.



**PARTS TO BE
ASSEMBLED**

YOUR PARTS

1 Servo motor



SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Servo cable (Stage 20)



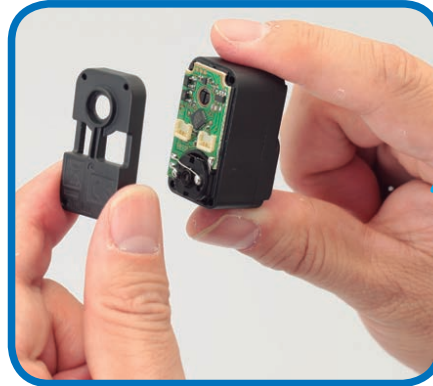
Neck stand (Stage 8)

THE SERVO AND CABLE

TIP!

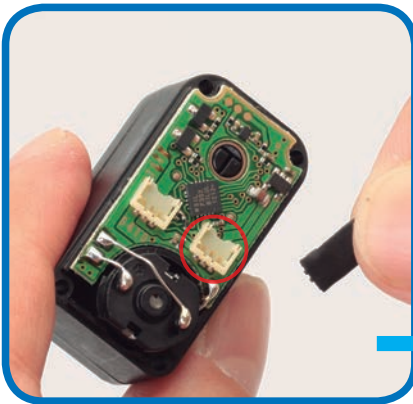


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.



TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with the pad in place, it is possible to apply the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.



4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back in place with all four screws, but only lightly as you will be removing them again at a later stage.

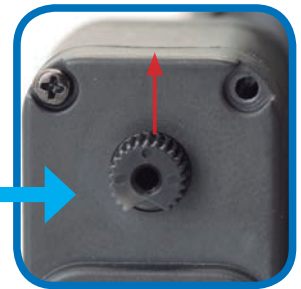
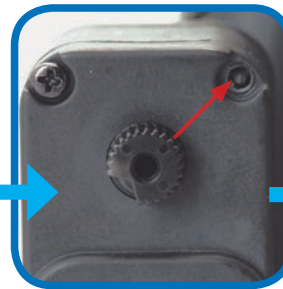
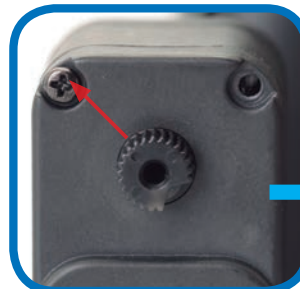
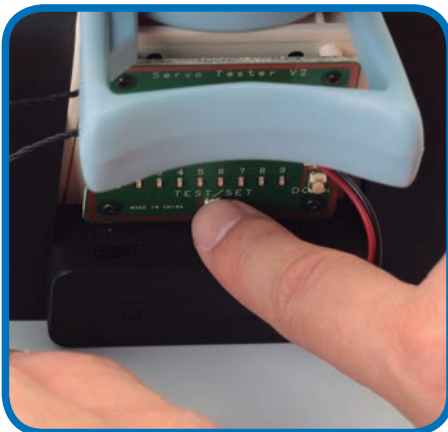
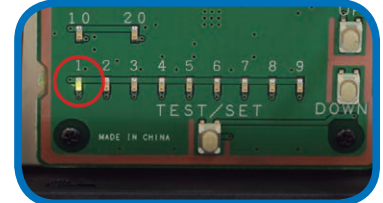
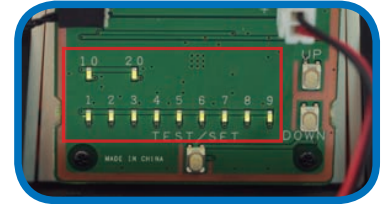
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 3.

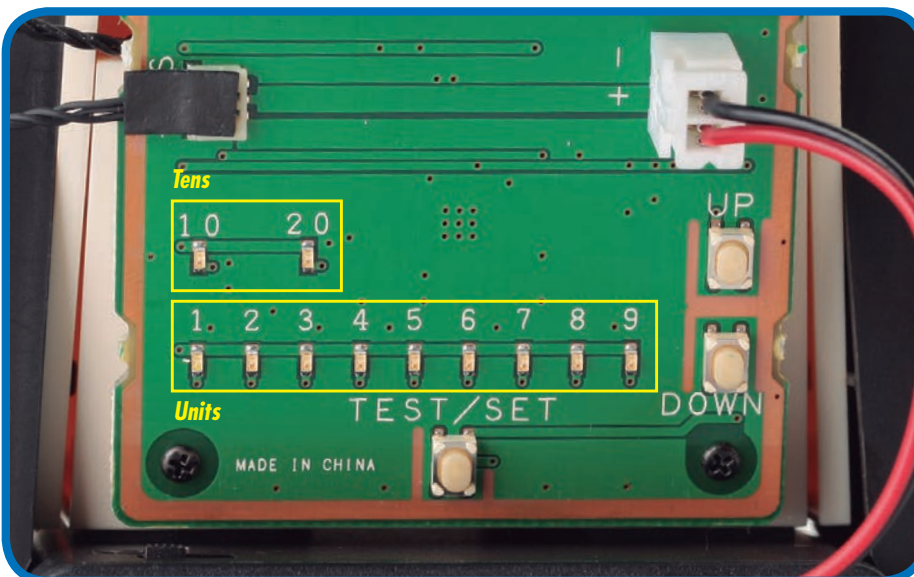


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

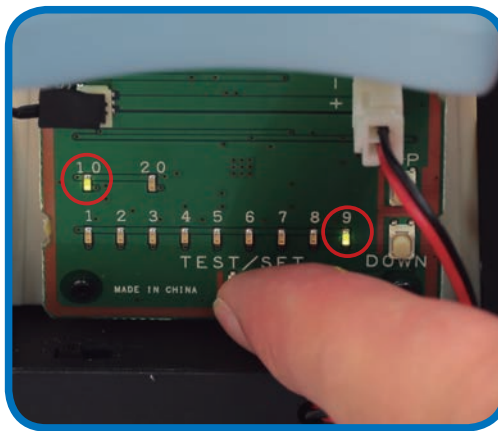
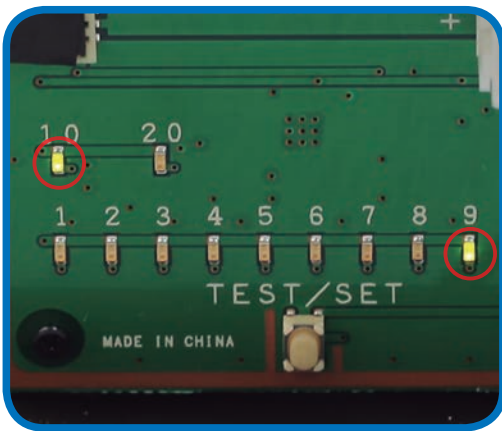


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

SETTING THE ID NUMBER



- 10** Re-familiarise yourself with the lights and switches on the test board. These are used to set the ID number of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 19, as you will do in this stage, you press UP 18 times. This will light up the 10 LED and the 9 LED. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.



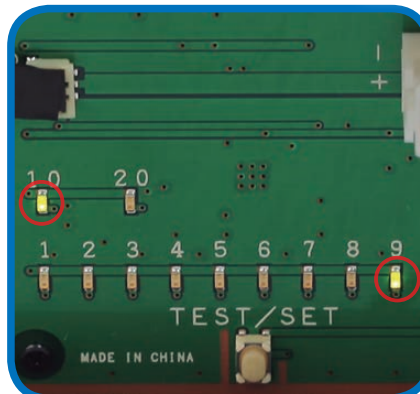
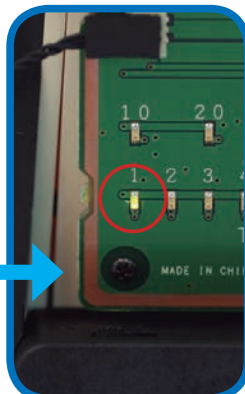
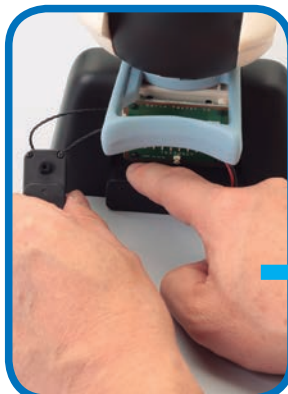
ID=19

11 To set the ID number of the left shoulder servo to 19, press the UP button 18 times. Make sure that the 10 and 9 LEDs are lit, using the UP and DOWN buttons to correct the number if necessary.

12 Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 10 and 9 lights should flash a few times, then remain lit.

CHECKING THE ID NUMBER

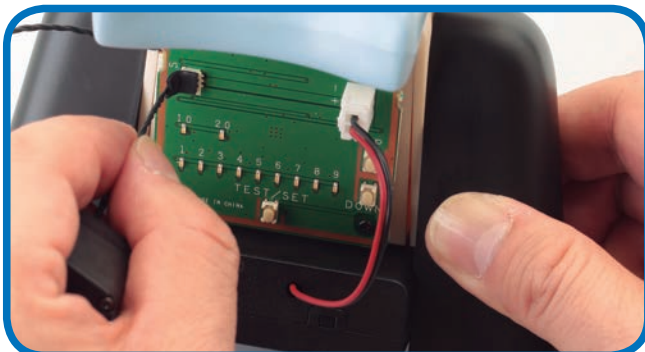
HELP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake or set the wrong ID number, you can reset it by going through the setting procedure again.

13 It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

14 To check the ID number, press TEST/SET. Now the 10 and 9 should light up. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.



Assembled shoulder servo

The servo that will operate Robi's left shoulder is assembled, wired and set. Make sure you keep this part safely, preferably in a sealed plastic bag with its ID number (19) written on it for ease of reference.



15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's shoulder.

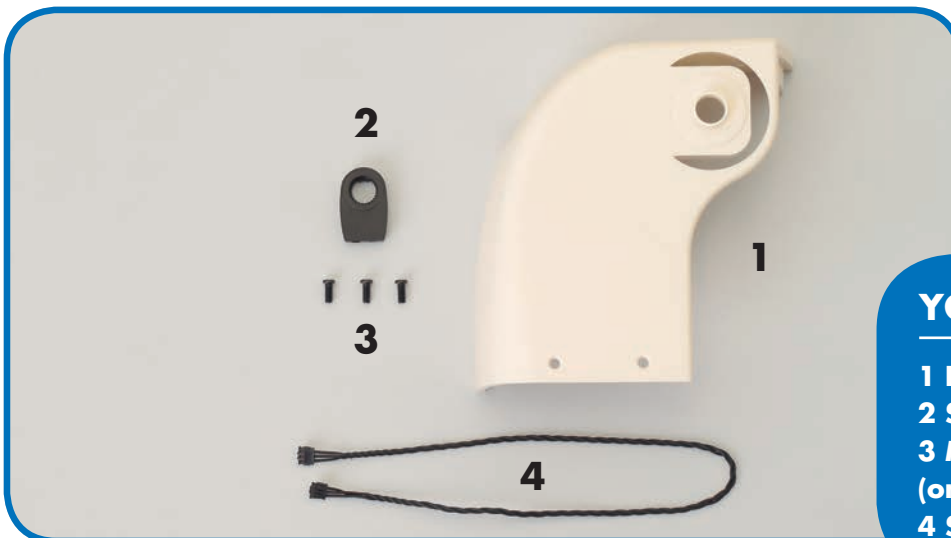
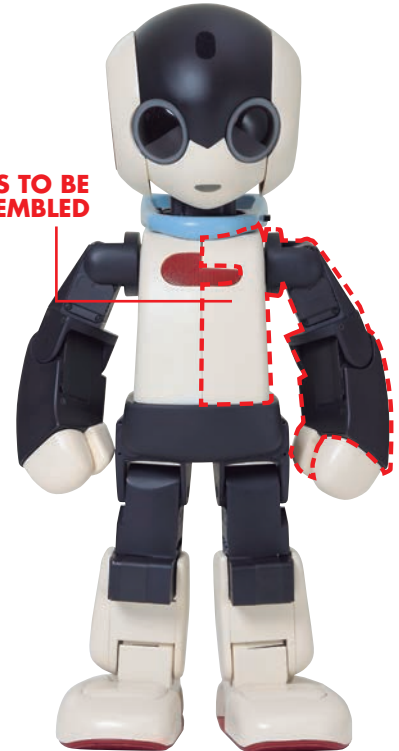
STAGE 22: CONTINUE BUILDING UP ROBI'S BODY

In this stage, you will continue assembling Robi's body, by attaching the left arm built and wired over the previous stages to his left body cover and shoulder servo.

This stage sees you working again on Robi's body, which you will do by fitting the previously assembled left arm onto the left body panel. You will then use a servo horn, as you did for the right arm, to attach the shoulder joint to Robi's left shoulder servo (ID=19), wired and

programmed in the previous stage. The part features a D-cut, and care must be taken to ensure the parts fit neatly. Once you have completed this, you will go on to wiring the first extended servo cable that will ultimately be used to operate Robi's right foot.

**PARTS TO BE
ASSEMBLED**



YOUR PARTS

- 1 Left body cover
- 2 Servo horn (shoulder)
- 3 M2 x 4.5mm screws x 3 (one is a spare)
- 4 Servo cable (200mm)

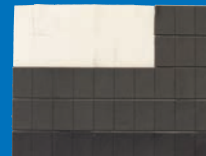
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



*Left arm assembly
(Stage 20)*

*Protective pad sheet
(Stage 3)*



*Wired servo ID=19
(Stage 21)*

THE RIGHT ARM AND BODY COVER



1 Carefully rotate the shoulder joint at the top of the left arm assembly so it sits at right angles to the rest of the arm.

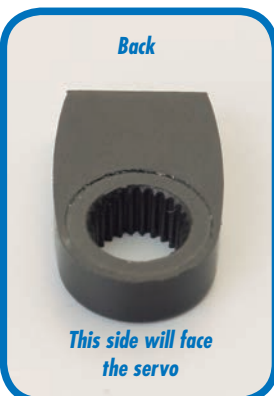


2 Line up the left arm to the left body cover.



3 Fit the servo cable emerging from the top of the left arm through the circular hole in the left body cover, and the metal shoulder joint through the C-shaped slot above it, as shown.

4 Join the parts so they sit together snugly and the cable and joint are fully inside the cover.



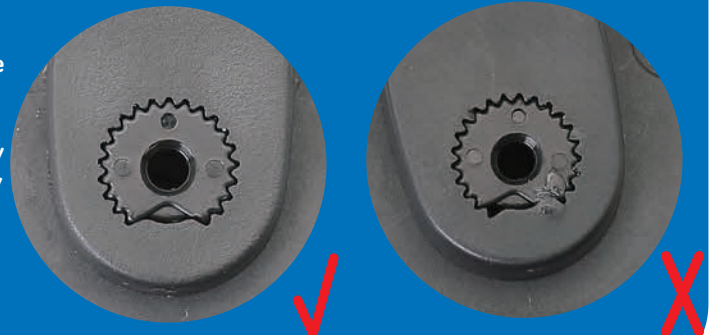
5 Inspect the servo horn: the hole in its front side has a D-cut profile, and will face outwards, while the back of the part has a round hole that will face the servo.

6 Take the servo from Stage 21 (ID=19) and line up the servo horn so that its front faces away from the servo. Then, with the servo horn perfectly straight with its D-cut section matching the cutaway on the servo's shaft, press the parts together. Do not force the parts, and see the Close Up box on the next page for more information.

THE D-CUT



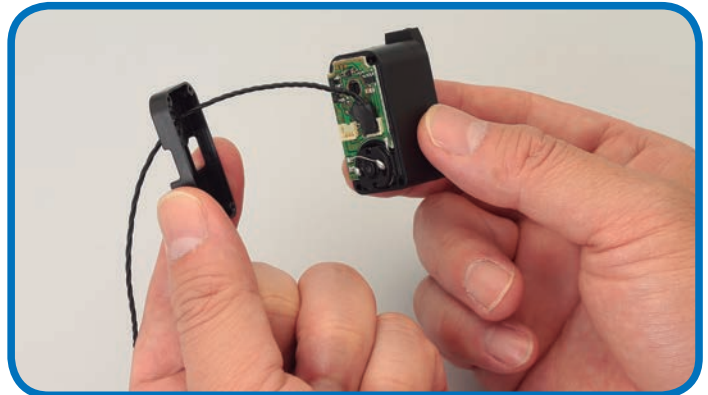
It is very important that you take care when working with the servos. Each joint, such as the servo horn in this stage, will have a specially moulded cutaway section known as the D-cut, which, when used properly, will fit snugly around the servo shaft to ensure it is set correctly. When these parts are aligned as they should be, it will be easy to slot them together. If it is in any way difficult to join the parts, remove them and check you have the D-cut section lined up to the shaft's cutaway and begin again. Never force the parts, as this could damage both of them.



FITTING THE LEFT SHOULDER SERVO



- 7** Remove the four long screws from the back of the servo to access the front panel.

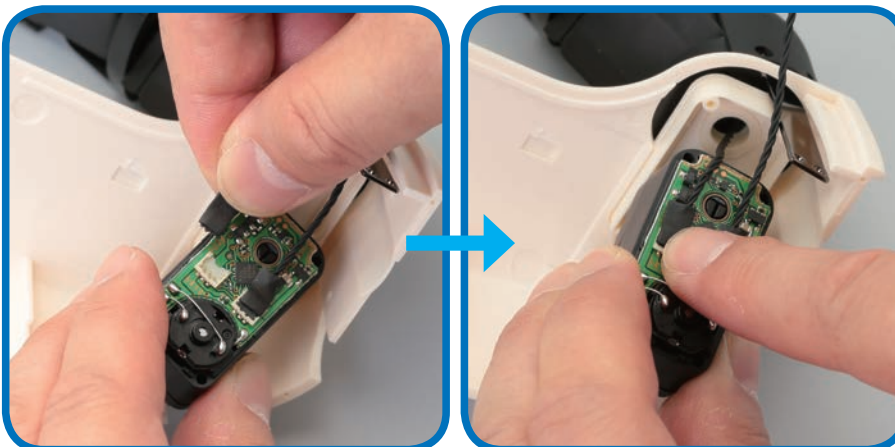


- 8** Remove the front panel, being careful not to pull the servo cable away as you do so. You will not be using this panel again.

TIP!



This servo cable had protective pads fitted to its connectors, but if you find it difficult to attach with the pad in place, it is possible to apply it afterwards. To do this, very carefully fit the connector into the servo's socket, then remove a pad from the backing sheet (Stage 3) and place on top of the connector and press down to seal the part.



- 9** Hold the open servo close to the inside of the left body panel, then very carefully press the free, padded connector of the left arm's servo cable into its free socket. This can be a tricky procedure, so work slowly and be sure not to pull on either cable, or touch the open electronics board of the servo.



Before sealing the servo, the cables need to be arranged in such a way that they do not become tangled. To do this, gently bend the cable leading into the arm out to the side of the servo's body, and the other back on itself and then out over the length of the part as shown. Be very careful not to disrupt either connector as you do this.



- 10** With the servo cable bent over and around to run into the notch in the body cover (circled, also see box to the left), press the servo's body onto the correspondingly shaped mount. Make sure the cable is not trapped or caught in any way.



As you tighten the screws, check that the servo cable and metal shoulder joint are still in position on the outside of the body cover.



- 11** Hold the parts in place, then using the four screws removed in Step 7, fasten the servo to the left body cover.



- 12** Remove the two screws on the shoulder joint and remove the protective panel.

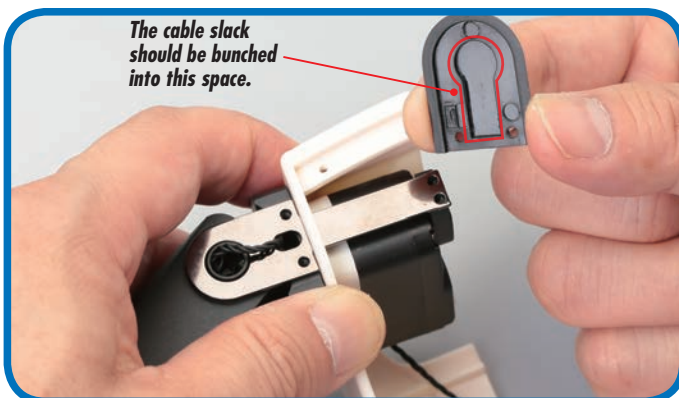
- 13** Very gently pull the exposed portion of the servo cable out from the elongated hole in the metal joint. Do not pinch the cable.



Eliminating the slack

The aim of this step is to remove any slack in the servo cable. Any loose sections along the servo cables set within Robi's moving parts, in this case the shoulder joint, are at risk of rubbing, or getting caught or pinched. This can ultimately damage the cables, and even cause Robi to malfunction, so be sure to follow these steps carefully and ensure that all the cables are set appropriately within the joints.

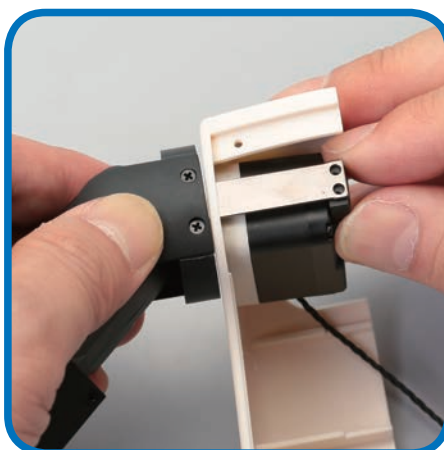
- 14** Lift the cable away from the side of the body panel so it is taut, then press down with your finger over the circular hole in the arm to prevent the cable coming loose.



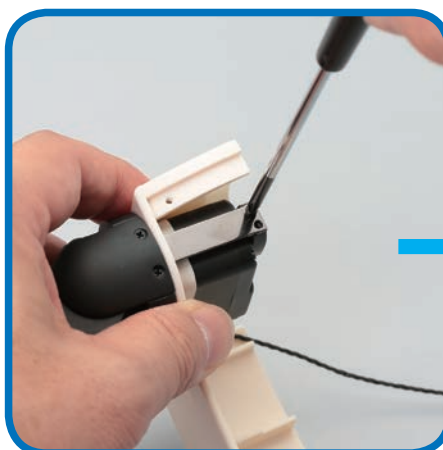
- 15** Keeping the cable tight to the body side, bunch the coiled section into a circle above the hole into the upper arm frame, to match the shaped recess in the inside of the shoulder panel removed in Step 12.



- 16** Fit the bunch of cable inside the recess, then, making sure not to pinch any part of the cable, replace the shoulder panel and tighten with its original screws.

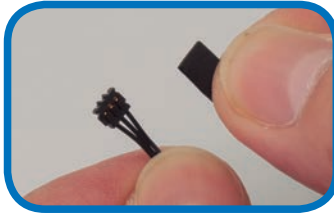
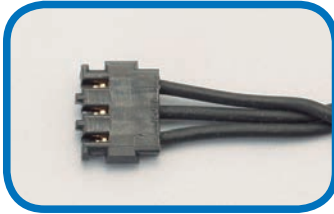


- 17** Line up the two holes at the end of the metal shoulder joint with those on the servo horn.

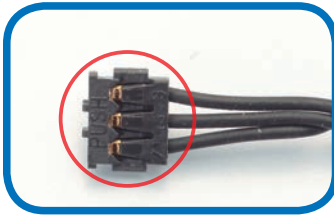


- 18** Insert and tighten an M2 x 4.5mm screw into each hole to secure the parts.

SEALING THE SERVO CABLE



It is advised that you fit the servo cable's protective pads at this stage, but if you want to wait until the cable is used in the build, you can. In this case, simply store the cable away safely until it is used, remembering to apply the pad then.



- 19** As you have done previously, apply a protective pad to the side of this stage's longer, 200mm servo cable with PUSH marked on it (circled). Repeat for both ends of the cable.

THE SERVO SCREWS



It is vitally important that the servo that moves Robi's shoulder is not hampered in any way, as this may cause a malfunction. Therefore, be sure to check that the screws on either side of the servo horn that connects to Robi's shoulder joint are fully inserted and do not obstruct the part's movement.



The servo horn will move back and forth, over the screws circled in the photo to the far left. Double-check that these are tightened fully into their holes so as to allow the servo horn freedom of movement.

To check that the servo horn is able to move easily, hold the body panel in one hand and with the other, slowly rotate Robi's right arm so that the servo horn covers each of the screws. If you feel any resistance, tighten the screw further and repeat until the arm is able to move freely.

Assembled left arm and body cover

The servos that will move Robi's left arm are now wired into the one hidden within his body to control his shoulder movements. Store your assembly away safely until next time, when you will start working on Robi's lower body.



COMING IN PACK 7

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 23

Prepare another of Robi's servos

PARTS PROVIDED

The first of Robi's ankle servos.



Stage 24

Start assembling Robi's right foot

PARTS PROVIDED

The first parts of Robi's feet – his right ankle covers.



Stage 25

Build up Robi's right ankle

PARTS PROVIDED

Robi's right ankle frame, tape and another servo cable.



Stage 26

Prepare the next servo

PARTS PROVIDED

The next servo that will move Robi's right ankle.